# Field Technician Operations Analysis

Natural Gas Distribution Company in Colombia | Dec 2025 - Feb 2026

## Key Performance Indicators

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|---|---|
| Total Service Orders | 7,643 |
| Overall Effectiveness Rate | 66.15% |
| Effective Orders | 5,056 |
| Non-Effective Orders | 2,587 |
| Planning Centers | 2 (Bogota / Cundinamarca-Boyaca) |
| Field Technicians | 40 |

## Executive Summary

Analysis of 7,643 field service orders from a natural gas distribution company in Colombia reveals a global effectiveness rate of 66.15%. The Cundinamarca/Boyaca center (2401) significantly outperforms Bogota (1501) with 75.49% vs 62.83% effectiveness. A steady decline was observed from December 2025 (67.4%) to February 2026 (63.4%), driven by operational process changes. Two key findings distinguish systemic failures from operational ones: service orders triggered by consumption deviations and uncoordinated scheduling by the distribution company account for a significant portion of non-effectiveness, which cannot be attributed to field team performance.

# Director of Operations Perspective

### Finding 1 - Effectiveness by Planning Center

Center 2401 (Cundinamarca/Boyaca) outperforms Center 1501 (Bogota) by 12.7 percentage points (75.49% vs 62.83%). Cultural and security factors in Bogota generate higher access refusal rates from users, which is a systemic issue beyond field team control.

### Finding 2 - Technician Performance

Significant dispersion exists in Center 1501 (43% to 82%) vs Center 2401 (63% to 87%). High volume does not determine effectiveness - Milton Alarcon leads volume (560 orders) with 75.4% effectiveness, while lower-volume technicians show worse rates, suggesting individual management issues.

### Finding 3 - Service Type Performance

Service orders triggered by consumption deviations (high-volume service type) show 52.8% effectiveness because users are not notified in advance. This is a systemic failure from the distribution company scheduling process, not a field team issue.

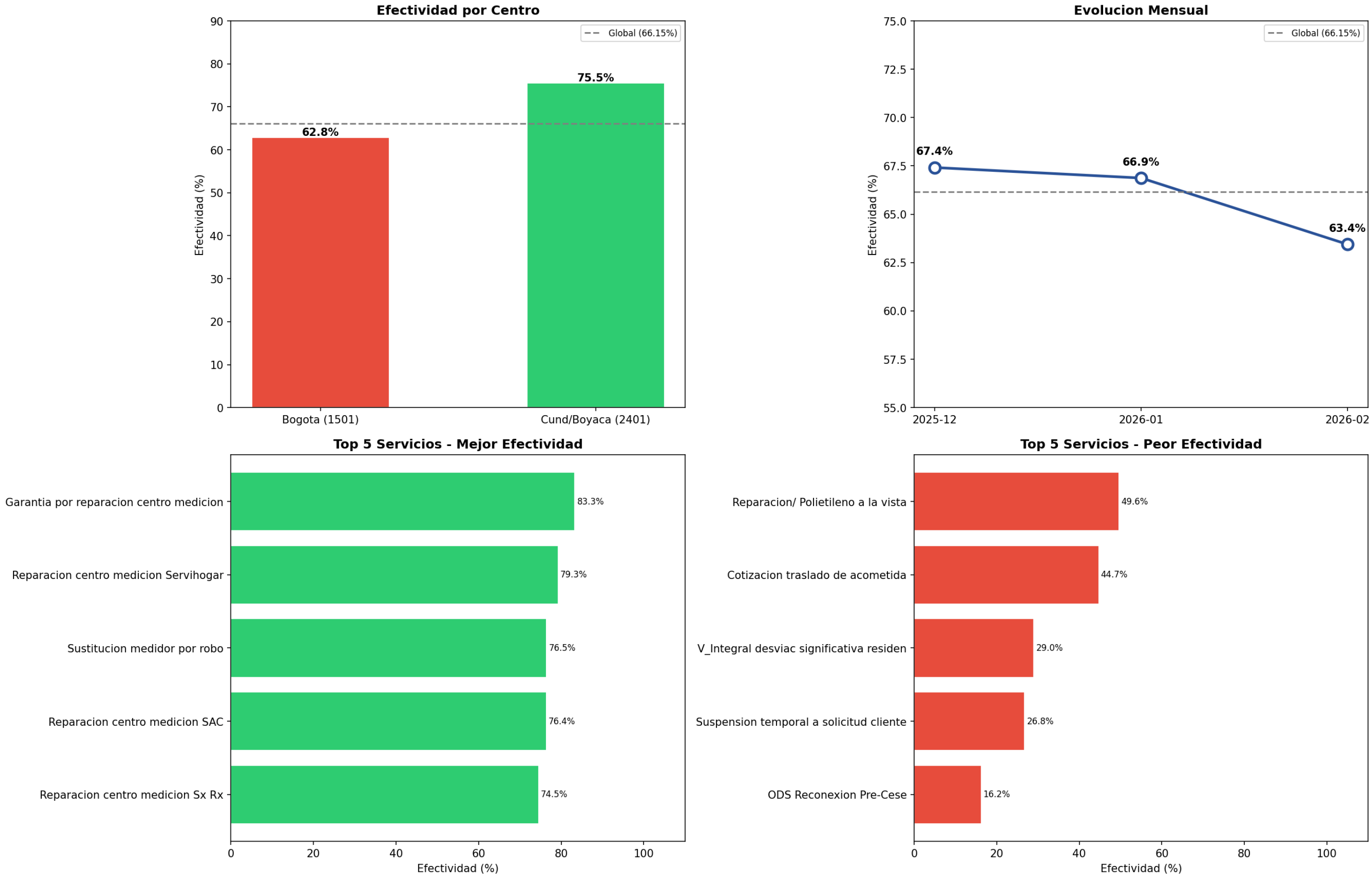
### Finding 4 - Monthly Trend

Effectiveness declined steadily: Dec 2025 (67.4%) - Jan 2026 (66.9%) - Feb 2026 (63.4%). The 4-point drop over 3 months correlates with operational process changes introduced by the distribution company during this period.

### Operations Dashboard

# Gas Operations Field BI - Executive Report

## Gas Operations Field BI - Executive Dashboard



## General Manager Perspective

### Finding 5 - Municipality Performance

Bogota concentrates 2,018 non-effective orders (62.46% effectiveness) - the worst rate among high-volume municipalities. Rural municipalities consistently outperform Bogota (Duitama 89.3%, Facatativa 88.8%), confirming the territorial factor hypothesis related to user trust and access.

### Finding 6 - Market Segment Performance

Industrial Non-Regulated leads with 83.0% effectiveness due to proper scheduling. Industrial Regulated shows the worst rate (42.1%) due to last-minute cancellations. Residential Nueva Edificacion (56.1%) presents the greatest opportunity for improvement among high-volume segments.

### Finding 7 - Service Type vs Systemic Issues

Key distinction: non-effectiveness falls into two categories. Operational (manageable by field team): repair and maintenance services. Systemic (caused by distribution company processes): pre-suspension reconnections (16.2%), temporary suspensions (26.8%), and deviation visits (29.0%) where users are not notified.

### Finding 8 - Technician Ranking

Rankings segmented by planning center reveal that Center 2401 is more homogeneous (24-point spread) while Center 1501 shows greater dispersion (38-point spread). Volume alone does not predict effectiveness - technicians with 300+ orders show rates from 43% to 75%, indicating individual management capacity differences.

## Strategic Recommendations

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| **IMMEDIATE** | 1. Revert operational process changes introduced in Dec 2025 - Feb 2026 that caused the 4-point effectiveness decline. Identify specific changes and their impact per service type. |
| **IMMEDIATE** | 2. Classify non-effectiveness by cause: systemic (distribution company) vs operational (field team). This separation is essential for accurate performance measurement. |
| **SHORT TERM** | 3. Implement user pre-notification protocol for deviation visits. Users must be contacted 24-48 hours before the visit to reduce the 52.8% non-effectiveness rate in this service type. |
| **SHORT TERM** | 4. Design individual improvement plans for technicians below 55% effectiveness in Center 1501. Focus on technicians with high volume and low effectiveness. |
| **MEDIUM TERM** | 5. Develop a differentiated territorial strategy for Bogota: visible technician identification, same-day confirmation calls, and immediate rescheduling protocols to address user trust barriers. |

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